



Engineer to Excel

ENERGY CONSUMPTION ANALYSIS

Using Statistical Estimation Techniques



Name : B C Yugesh

Reg no: 192424282

Course: Fundamentals of Data Science

Course code: DSA0404



WHAT IS IT?

Statistical estimation uses sample energy data to estimate the typical consumption of a larger population. It helps convert daily readings into useful evidence for planning, budgeting and conservation.

OBJECTIVES

- Estimate the average daily energy requirement.
- Measure variation in consumption using sample variance.
- Construct a confidence interval for the population mean.
- Detect unusually high-use periods.
- Support energy-saving and resource-planning decisions.

SAMPLE DATASET

Day	Energy (kWh)
Mon	118
Tue	132
Wed	125
Thu	148
Fri	155
Sat	138
Sun	144

KEY CONCEPTS

POINT ESTIMATION

$$\hat{\mu} = \bar{x}$$

A sample mean is used as the best single estimate of the population mean.

INTERVAL ESTIMATION

$$\bar{x} \pm t(s/\sqrt{n})$$

Gives a plausible range for the true average.

STATISTICAL ESTIMATION

Sample size (n) = 7

Total consumption (Σx) = 960 kWh

Estimated mean ($\bar{x}$) = 137.14 kWh

Sample variance (s^2) = 170.81 kWh²

Sample standard deviation (s) = 13.07 kWh

95% CI = $\bar{x} \pm t(s/\sqrt{n})$, t = 2.447

95% CI = (125.06, 149.23) kWh

MAIN RESULTS

**ESTIMATED
MEAN**

137.14 kWh

**95% CONFIDENCE
INTERVAL**

125.06 – 149.23

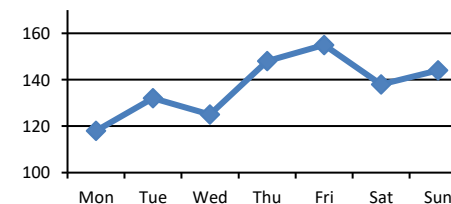
INSIGHTS FROM ESTIMATION

- Average estimated consumption is 137.14 kWh/day.
- The 95% confidence interval ranges from 125.06 to 149.23 kWh/day.
- Friday records the highest usage at 155 kWh.
- Monday has the lowest usage at 118 kWh.
- Variation is measurable but not extreme, suggesting a fairly stable weekly pattern.

PRACTICAL APPLICATIONS

- Forecasting demand
- Budgeting electricity costs
- Identifying peak-use periods
- Evaluating energy-efficiency programs
- Supporting sustainability targets

Daily Consumption



Usage Distribution

